

A Virtual Testbed for Security Management of Industrial Control Systems

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Abstract—Industrial Control Systems (ICS), at the heart of critical infrastructures, have been developing into more and more powerful tools over the years with the incorporation of networking and cyber, among other, technologies. As with great power comes great risks, ICS's inherited risks and vulnerabilities not only from the cyber domain but also gave rise to unique security risks and vulnerabilities. Security management, i.e., identifying vulnerabilities, assessing threats, preventing and tolerating malicious and harmful activities, mitigating and recovering from attacks, etc., for ICS's have been identified as a key, if not the key, area for the protection of critical infrastructures. The importance of a test-bed mimicking a real-life ICS and having capabilities to support ICS security management is paramount. In this paper, after briefly surveying the vulnerabilities in ICS and surveying the existing ICS test-beds, we describe the implementation of a virtual ICS testbed controlling the distribution breaker system of a power grid. We also describe simulating two cyber attacks using the testbed.

I. INTRODUCTION

Industrial Control System (ICS) is an umbrella term, which alludes to a collection of various control sub-systems; commonly including: a) the Supervisory Control and Data Acquisition (SCADA) systems, b) Distributed Control Systems (DCS), and c) Programmable Logic Controllers (PLC) [1]. This control system is then used to operate physical devices and components including field devices, sensor systems, control panels, etc. in many critical infrastructures like electrical, water and wastewater, oil and natural gas, and industries like chemical, transportation, pharmaceutical, pulp and paper, food and beverage, discrete manufacturing (e.g., automotive, aerospace, and durable goods), and others [2]. A typical ICS architecture is shown in Figure I.

A particular ICS may comprise of just an SCADA system or a combination of these components depending on the size, complexity and architecture of the industrial process it monitors and controls. The ICS in this paper comprises of SCADA system, a PLC and a distribution substation of a power grid.

Industrial Control Systems and its various components have been undergoing evolution for decades. However, the most significant changes came when these self-contained and isolated systems have been empowered with networking capabilities, particularly when ICS's are connected to organization's IT networks in order to enable managers and operators the ability to monitor and control them from their desktops and other consoles. This essentially rendered ICS's as cyber-physical systems (CPS) with the power and flexibility of CPS. However,

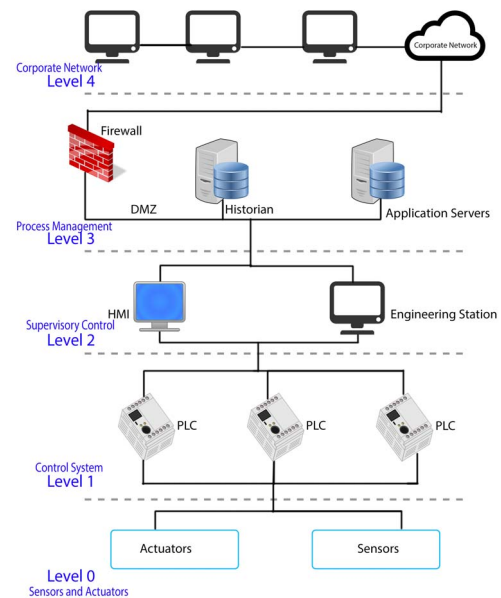


Fig. 1. General layout of a ICS Architecture.

it also rendered the so-called “air-gap” between the IT and OT networks ineffective [3], allowing the cyber-security issues to creep into the CPS [4].

Investigation of security ramifications in ICS's has recently become a very important research area [5]. Even though the main thrust of this research is the protection of critical infrastructures. Because of the large and cumbersome nature of, not to mention exorbitant cost, it is not possible to use a real-life and operating ICS for security management research. In that place we need a software-based virtual testbed. This paper describes the first phase of implementation of a virtual testbed for power grid distribution station emulated using MATLAB Simulink and PLC Simulator.

The remaining part of this paper is organized as follows. Section II surveys a few ICS testbeds currently available. Section III describes the architecture and implementation of the emulated Power Lab Testbed. Section IV describes the experimental setup and results. Section V describes the simulation of two cyber-attacks using the emulated testbed,

and finally Section VI concludes this paper.

II. RELATED WORK

The Virtual Power System Testbed (VPST), at the University of Illinois at Urbana-Champaign, was designed to be integrated with other testbeds; in order to study more about the SCADA equipment, its protocols' security, and performance [6]. The network simulation in VPST is based on RINSE, a network simulator [7], which is used to model the cyber domain of an electric grid. Electric simulation is handled by "Power World" simulation utility in order to simulate the large scale electric networks.

The SCADA Security Laboratory and the Power & Energy Research Laboratory, at the Mississippi State University, consists of multiple critical infrastructure components; in the form of: commercial hardware and software modules, along with common industrial control system networks [8]. This testbed can be used to identify different security vulnerabilities of controlled processes. Varied mitigation techniques can be devised and verified against the security vulnerabilities found. There are several physical devices being utilized in this testbed. Different protocols, including GOOSE, MODBUS, and DNP3, are supported by this testbed [9].

Virtual Control System Environment (VCSE) was developed by Sandia National Laboratory, in order to discover and address the vulnerabilities of targeted attacks on energy systems. This testbed contains Simulated, Emulated, and Physical components for Investigative Analysis (SEPIA) [10]. VCSE is not just a tool, but an environment where multiple tools are put together; in order to study various control systems. Simulation models could be built, saved, and restored during any point of the simulation process. This testbed provides a secure, and encrypted channel of data communication on the IP network.

The Testbed for Analyzing Security of SCADA Control System (TASSCS) was developed to protect SCADA systems from cyber attacks using techniques like false alerts, detection rate etc. [11]. The primary objective is to evaluate the Autonomic Software Protection System (ASPS).

The Intrusion and Defense Testbed was developed at the University of Dublin. Anomalies and intrusions can be detected in this testbed [12]. Using the information, and the data that has been acquired at various points of time; temporal anomaly can be determined - which is used to determine an anomaly amongst different time stamps. The architecture comprises of 2 substations and 2 control centres. Local Area Networks (LANs) using various industrial protocols are used for connection among different subsystems. The testbed also provides an external link to the Iowa State University testbed.

III. VIRTUAL TESTBED ARCHITECTURE

Our virtual testbed architecture consists of two subsystems: i) the SCADA subsystem is emulated using the MATLAB Simulink and ii) the PLC subsystem [13] is emulated using the Modbus PLC protocol simulator [14] and the MATLAB Simulink [15].

The Virtual Testbed consists of 3 different Virtual Machines. The first VM is used to simulate the SCADA system. The second VM is used to simulate PLC which runs the control logic and the third VM is the attacker VM. We need to see that all the three VM's are on the same network. Figure 2 shows the architecture of the testbed.

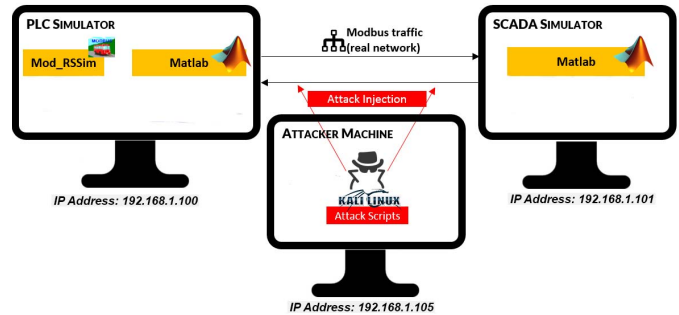


Fig. 2. The Virtual Testbed Setup.

A. Subsystem 1: The SCADA Emulation

Figure 3 shows the physical system emulated as part of the SCADA system in the testbed. The physical system is a component of the Power Lab in the Electrical and Computer Engineering department at the University of Idaho that mimics a distribution substation of a power grid. The distribution system consists of two 3-phase sources and two loads. The sources are connected to a "VI" Measurement block (Voltage and Current measurement block) each. Next to the VI measurement blocks are the Three-Phase breakers followed by the Three-Phase Pi Section Lines. After the Pi Section lines, there are two different bus systems: one on the top and the other on the bottom. The top and the bottom bus lines there are similar blocks and both are mirror images. In between the two Pi-section lines that are present next to each other we insert the "Load" since the model follows a "Load Shedding Scheme".

Figure 4 shows the MATLAB diagram of the emulated SCADA system. Utilization of Simulink is done for the purpose of model-based design, where different block environments are used. Model-based design is used to develop different dynamic systems like control systems, signal processing, and communications systems. Once a required system is developed, the simulation enables us to verify whether the model works properly or not. There are several toolboxes that can be plugged into Simulink, so as to model our system. The designed testbed consists of PLC model, as well as a SCADA system model; both of which have been modeled in MATLAB Simulink. In addition to the default toolboxes of Simulink, we use two other plug-ins.

- **Instrument Control Toolbox:** Instrument Control toolbox allows us to connect different instruments using: a) the different instrument drivers or b) the specific communication protocol being utilized; for example: UDP, TCP/IP etc. Using this toolbox, we can generate data sets in MATLAB. These data sets can be then used for: a)

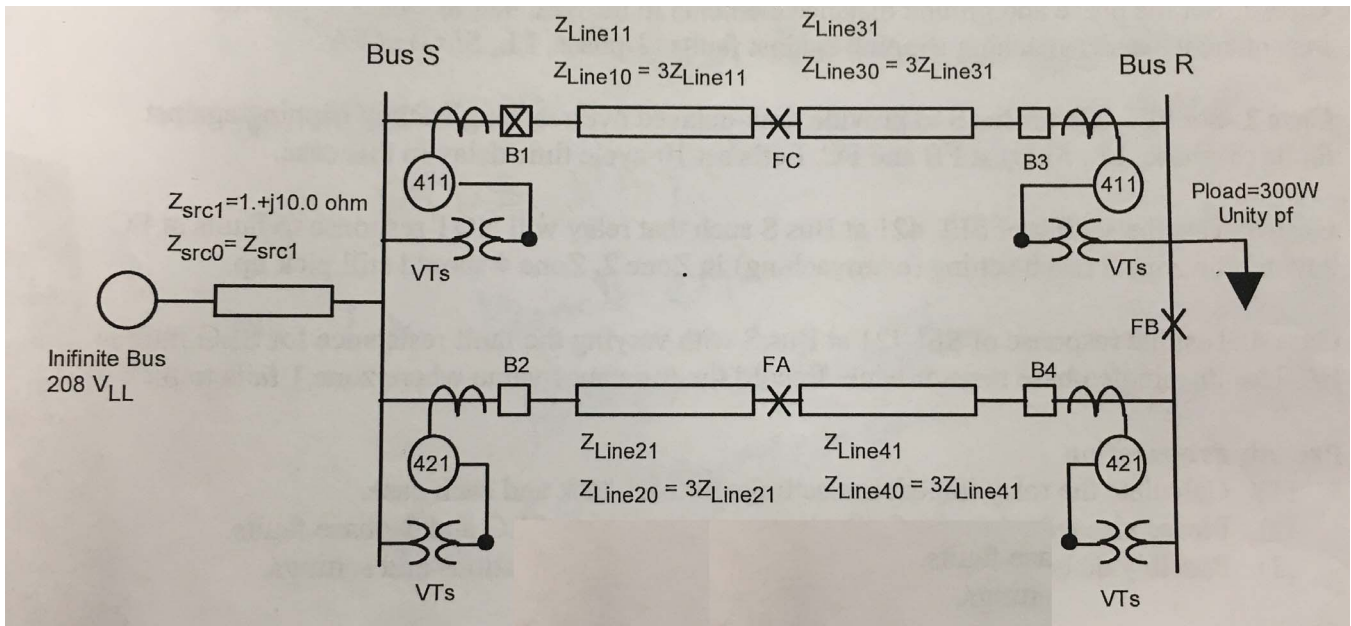


Fig. 3. The Distribution Substation Component in the ECE UI Power Lab.

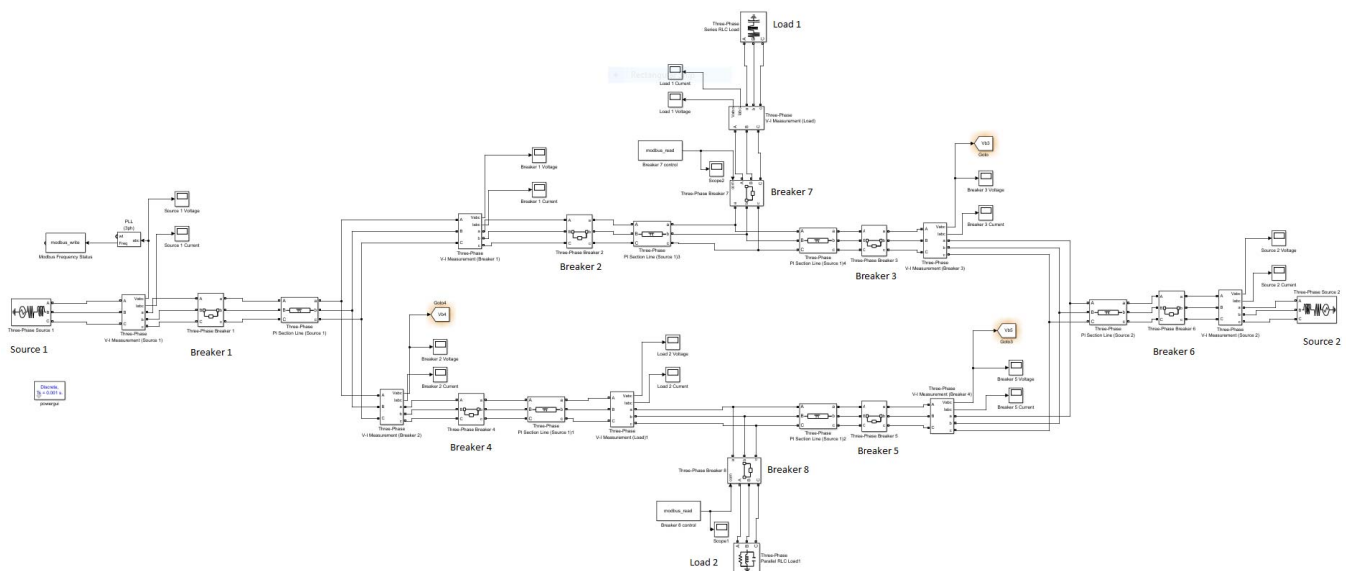


Fig. 4. The SCADA System Emulation of Distribution Substation Component using MATLAB Simulink.

data dissemination by exporting, and/or b) further data analysis within MATLAB itself. Remote communication with other computers can also be possible using this toolbox since it supports TP/IP, UDP, Bluetooth, and other such protocols [16].

- **Simscape Power Systems:** Simscape Power Systems provides the users with component libraries, which are used to model, develop, and simulate electrical power systems. This toolbox also enables the users to test the system-level performance. The library includes almost all

of the power components including: three-phase power modules, and renewable energy systems [17].

B. Subsystem 2: The PLC Emulation

Modbus was first introduced in 1979. It was designed by Modicon company, and is one of the most used protocols for communication in Industrial Control Systems[18]. Modbus is mostly implemented over TCP. Specifically, it operates at the 7th layer of OSI model, and has no form of native authentication mechanism. It is normally used for communication

between SCADA and PLC. The device which is requesting the information is called “*Master*” and the device that is providing the information is called “*Slave*”. In our architecture, the SCADA system is the *Master* and the PLC is our *Slave*.

Modbus PLC protocol simulator [14] was used to simulate multiple Modbus slaves with proper register configuration. Using this configuration, we can respond to the Modbus/TCP messages which are sent by the Modbus Write block. The information is stored in the coils. Typically, registers can be divided into two categories: a) holding registers - which are used for configuration values, and b) input registers - which are used for status and measurement. Each register can hold up to 16 bits. The protocol used for communication between PLC and SCADA system is Modbus/TCP.

Similar to the SCADA modeling, there are several blocks that we have utilized in the process of modeling a PLC, to virtually simulate the working of a PLC. Apart from the toolboxes available by default in MATLAB Simulink; the process of designing a PLC utilized two toolboxes: 1) Simscape Power Systems, which is a part of Simscape Toolbox; and 2) the Instrument Control Toolbox. A few blocks are similar to what has already been used in the SCADA system modeling, such as the “Modbus Read” block and the “Modbus Write” block.

C. Data Acquisition

In our SCADA emulation we put “scopes” to read physical parameter values, like voltage and current, off the bus. MATLAB allows the scopes to be moved around and placed at different points. When running a simulation MATLAB exports all the data read by the scopes in a useful format, e.g., CSV. As there are six scopes, we collect data from six different points while running any simulation experiment. The data is collected for every microsecond. Each data collection point has six different values. Three values for voltage and the other three for current. Since we implemented a 3-phase circuit we get the value for each of the phases ‘a’, ‘b’ and ‘c’ for both voltage and current.

IV. EXPERIMENTS AND RESULTS

In this section, we demonstrate the functionality of our testbed when operated under different breaker and PLC conditions. We also perform reconnaissance attack and denial of service attack on the communication channels of our testbed and discuss the results of these respective attacks.

A. Experimental Setup

The experimental setup consists of 3 different Virtual Machines (VM). First VM runs the emulated SCADA system described in III-A. Second VM runs the emulated PLC described in III-B. The third VM is the attacker VM. All three VMs are mutually connected through the same network. The SCADA and PLC VMs run Microsoft Windows 10 and the attacker VM runs Kali Linux.

B. Breaker Conditions

The Power Lab at University of Idaho follows a load shedding scheme (two breaker conditions), and since we are trying to remodel the power lab design, our scheme will also have two breaker conditions. They are: a) Normal working condition (breakers closed), and b) Breakers open condition.

1) *Normal Working Condition (Breakers Closed)*: Whenever the simulation is executed; power is transmitted to all the available loads. Voltage values are calculated once every microsecond, for the entire duration of the simulation. These values are then sent to the PLC model through “*Modbus Write*” blocks, wherein the voltage values are written in the registry. “*Modbus Read*” block in the PLC model then reads voltage values from the registry, and performs the necessary calculations. The calculations include getting the voltage value, calculating the root mean square of it, followed by the mean all the root mean square values. It is then compared to a constant and then send to a logical operator depending on the condition. These calculations lead to the generation of a breaker signal, which specifies whether to open, or close the breakers; depending upon the conditions specified.

In a normal working condition, the total voltage without any loads attached to the circuit should be nearly 180 volts. But when we introduce loads, there will be a difference in voltage, since the loads will absorb some voltage. Hence, depending upon values of our loads, voltage across our circuit will vary. When there is no breaker open, we observe that the voltage plot of our circuit results in a sinusoidal waveform. At any given point of time the sum of the voltage values of all the three phases will be equal to zero. The waveform will be same across all the different scopes in the circuit, since there are no breakers open.

2) *Breakers Open Condition*: In a condition where the breakers are opened; the circuit becomes an open one, and there will be some disturbances in the circuit’s voltage waveform. Upon opening of the breakers, waveform tends to get disturbed, resulting in a diversion from the sinusoidal waveform.

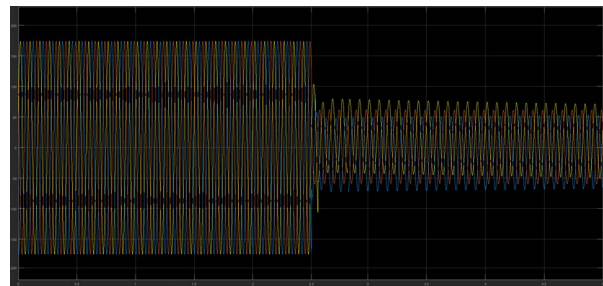


Fig. 5. Waveform Showing the Breaker-Open Condition.

From the Figure 5, we can see that the breakers remained closed up to a certain point, and the voltage is represented with a sinusoidal waveform. When breakers are opened, we see a sudden drop in voltage and at this point, the voltage waveform is no longer sinusoidal.

C. PLC Conditions

PLC contains a set of conditions which, when satisfied, produce a value depending on the condition. Physical PLCs work on Ladder Logic programs. However, in our model as discussed in III-A, the PLC is emulated using Simulink blocks. There are a total of 7 different conditions that are implemented in the PLC model. They are:

- 1) If voltage at Breaker 3 is less than 80%, then open Breaker 7.
- 2) If voltage at Breaker 8 is less than 80%, then open Breaker 8.
- 3) If frequency at Breaker 1 is less than 58Hz, then close either Breaker 7 or Breaker 8.
- 4) If Breaker 6 is open, and Breaker 2 is open; and voltage at Breaker 4 is less than 80%, then trip Breaker 7.
- 5) If Breaker 6 is open, and Breaker 4 is open; and voltage at Breaker 4 is less than 80%, then trip Breaker 8.
- 6) If Breaker 1 is open, and Breaker 3 is open; and voltage at Breaker 4 is less than 80%, then trip Breaker 7.
- 7) If Breaker 1 is open, and Breaker 5 is open; and voltage at Breaker 3 is less than 80%, then trip Breaker 8.

V. SIMULATION OF CYBER ATTACKS

In the contemporary practice, ICSs use a polling approach to monitor the system continuously. Polling method takes in a request to the server from a client, and responds back. This response is evaluated for anomalies. Many of the network protocols used in ICSs lack authentication to validate the origin of a packet. This opens an avenue of attack, by allowing attackers monitor, capture, and edit a packet, which contains critical sensor values. In this section, we demonstrate how two common cyber-attacks, namely reconnaissance and DoS, can be simulated using our virtual testbed.

A. Reconnaissance Attack

Reconnaissance is a type of attack, where the attackers obtain information about the target system. This includes details along the lines of: type, manufacturer, model number, address of the system, and the different network protocols supported/utilized. This information can be used to prepare an actual attack on the target system. Following information is usually obtained, to increase the chances of an attack's success:

- 1) Devices available on a network, and types of the devices;
- 2) Protocols available on a network, and utilization ratio of the protocols;
- 3) Identify devices most likely to carry critical information and/or be able to perform critical operations;
- 4) Once such devices are identified, obtain the addresses of such critical devices.

Modbus/TCP protocol is confined to TCP port number 502. Since the Modbus protocol follows a client-server architecture; all the requests to a server are responded back, even if the request results in an error, which reveals existence of a Modbus device. We implemented this attack using an address scan,

where we send a request to the Modbus Device at a random IP address. This request asks the server to read the value of a particular holding register. If we receive any response, that might be an indication that a Modbus device exists at that specific address.

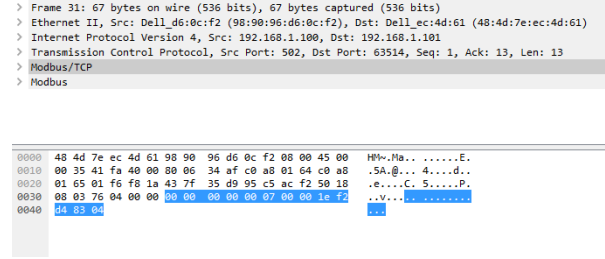


Fig. 6. Attack Response in the Modbus packet during a Reconnaissance Attack.

In Figure 6, the highlighted portion shows the Modbus/TCP part of the received response. Using the official Modbus messaging guide, we know that '83' implies a Modbus/TCP error, and '04' implies the exception code 4 - which means a server failure. This information indicates that there is no registered client with our current device number and hence some adversary might be snooping on the system.

B. Denial-of-Service Attack

Denial-of-Service (DoS) is a type of attack where a server is made unavailable to all the legitimate users of a service, by the attackers. An attacker can disrupt the network, preventing user-access to email, websites etc. The most common technique causing a DoS attack is: network flooding; of which, the following is an example: Accessing a URL will spawn a request, that is sent to the Server which hosts the URL's resource. This Server can respond to only a limited number of requests, at any given time. As such, if the attacker overloads said Server with requests at all times during an attack, then the Server drops any subsequent requests[19]. There are many other ways using which, an attacker can cause a denial of service. For example: disturbing communication between a client and the server; which prevents client's access to the service.

In our default testbed setup, there is an exchange of packets; which contain: a) the values of different involved testbed-components, and b) information that needs to be sent from the SCADA system to the PLC. We, then, launch a Denial-of-Service attack from the attacker VM; i.e., a Kali Linux system - using a pre-installed software application called *Ettercap*.

VI. CONCLUSIONS

In this paper we described the emulation of an ICS comprising of a SCADA with power distribution system and a PLC, implemented using MATLAB Simulink and PLC Simulator. We have also showed using two examples that this virtual Testbed is a useful tool for research in the field of cyber security of ICS. Use of Simulink model guarantees the fidelity of the emulation and also decreases the time needed to perform

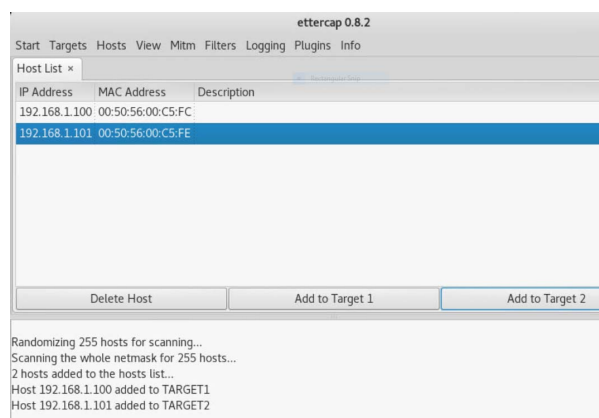


Fig. 7. Ettercap Hosts List

any modifications to the model, making it flexible and easily extensible.

The most important property of this testbed is its capability to generate and export different physical parameter values. The data exported can be treated for security analytics for quick and accurate detection of cyber-security breach in power-grids and use the insight gained for autonomic self-healing and assess economic impacts.

A. Future Work

The current work primarily focused on a SCADA system around the distribution substation in the Power Lab in University of Idaho. This can be extended to incorporate other components in the Lab and also model different other SCADA systems like the Gas Pipeline, Water Storage tank etc. The current model can be expanded to have multiple PLC's controlling the breakers rather than just the one PLC model controlling all the breakers in the current work.

Man in the Middle attack could be another type of attack that can be performed on the current model by modifying the packets on the fly and then sending in the wrong values to the breakers even though there is no fault and asking them to open which causes a disruption in the steady running state of the circuit.

Our next step is to perform security analytics on the data gathered from the testbed, using techniques including machine learning, deep learning and/or AI-based algorithms. This analytics can help us devise a technological solution to accurately ascertain the deviation from the "normal and identify if the deviation is due to activities of a cyber adversary. In addition, the insights gained from this analytics can be used to improve business continuity from the cybernomics point of view [20].

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